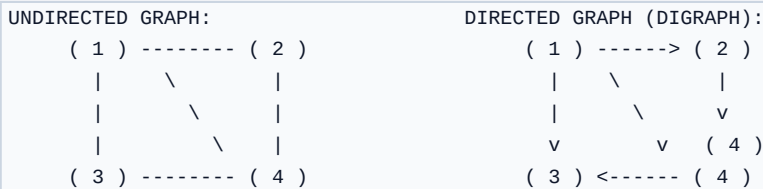


Graph Introduction & Terminologies -- Data Structures

Definition: A Graph is a non-linear data structure denoted as $G = (V, E)$, consisting of a non-empty finite set of Vertices (Nodes) V and a set of Edges (Arcs) E connecting pairs of vertices.

1. Detailed Technical Explanation



Core Graph Terminologies:

1. Vertex (Node): An individual data point or entity in the graph.

2. Ed

8. Adjacent Vertices (Neighbors): Two vertices connected directly by an edge.

9. Pa

2. Memory Keywords & Properties

* Vertices & Edges: $G = (V, E)$.

* Ha

3. Must-Write Points for Exams

* In an undirected complete graph with N vertices, the maximum number of edges is $N(N - 1) / 2$.

* In

4. Quick Recall Flow

$G = (V, E) \rightarrow$ Directed vs Undirected $\rightarrow$ In/Out Degree $\rightarrow$ Handshaking Lemma ($\text{Sum Deg} = 2E$) $\rightarrow$ Complete Graph $E = N(N-1)/2$